PROJECT PROPOSAL • EMBEDDED AI • FALL 2026

SWING TO MINE

A motion controller for Minecraft, with the neural network running on the chip.

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THE IDEA • BUILD TRACK

Hold the board like a sword. Swing it, and Minecraft swings.

It plugs in as a keyboard and mouse

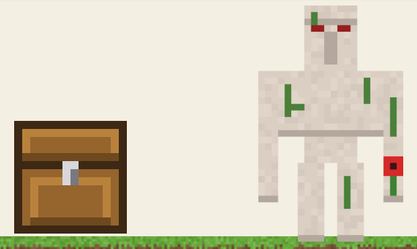
The board shows up over USB like any keyboard and mouse. Nothing to install on the PC.

The model runs on the chip

A small int8 neural network on the STM32F407 spots each gesture. No cloud and no PC.

It is the whole controller

A gyro handles looking around and tilt handles walking. No real keyboard or mouse needed.



HOW IT WILL WORK

From a swing to a click in four steps

STEP 1

Sense

A gyro and accelerometer read motion 100 times a second.

STEP 2

Classify

The model checks the last second of motion every 50 ms.

STEP 3

Decide

A few frames must agree before anything fires.

STEP 4

Send

The board sends the key press or click over USB.

Only gestures go through the model. Looking and walking use plain math on the gyro and the tilt.



THE GESTURES

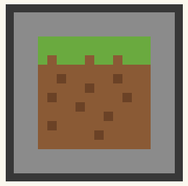
The moves the model will learn



LEFT CLICK

Attack, mine

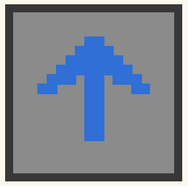
Slash or chop



RIGHT CLICK

Place block

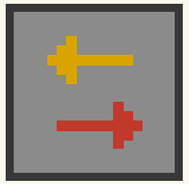
Punch forward



SPACE

Jump

Hop on the spot



SCROLL WHEEL

Swap item

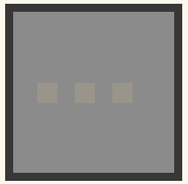
Twist your wrist



E KEY

Inventory

Shake the board



NO INPUT

Nothing

Any other motion



LOOKING AND WALKING

Looking and walking need no model

LOOK MODE

Point to aim

The gyro measures how far the board turns, and the camera turns with it. Aiming pauses during a swing.

MOVE MODE

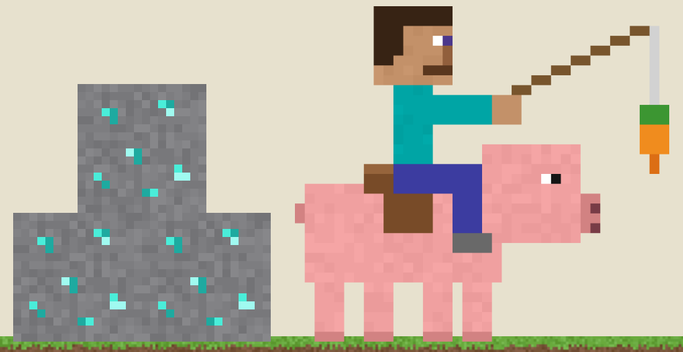
Tilt to walk

The board works like a joystick. Tilt forward to walk and sideways to strafe.

BLUE BUTTON

Switch modes

A short press switches look and move. A long press recalibrates. The LEDs show the mode.



THE DATA

We will record our own dataset

4-8

people, including both of us

40

reps of each gesture from each person

10 min

of “nothing” from each person

Why “nothing” matters most

The model watches live motion all the time, and most of it is not a gesture. This class is what stops accidental clicks.

How we will test it

One person’s data stays out of training completely. We test on that person, as if a stranger picked it up.

THE HARD PARTS

What makes this harder than Lab 2

A live stream, not clips

In Lab 2 each file held one gesture. Here the model has to find the gesture inside nonstop motion.

Mostly not gestures

Aiming, walking and fidgeting must all count as nothing, or the game fills with accidental clicks.

Fast and small

It has to feel instant and share the chip with the USB code: 1 MB of flash and 192 KB of RAM.



TARGETS, NOT RESULTS YET

How we will know it worked

90%

accuracy or better on a person the model has never seen

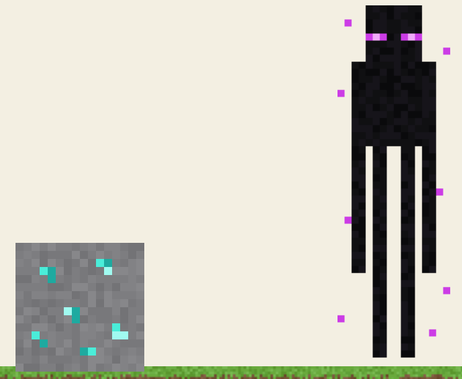
100 ms

or less from the motion to the input in the game

1/min

accidental gesture at most during normal play

We will measure all three on the real board and report the numbers, along with flash and RAM use.



THE PLAN

Working controller by Thanksgiving

NOV 12–15

Talk to the board

Wire the gyro and stream its motion data to a PC.

NOV 16–19

Controller, no model

USB keyboard and mouse, look and move. Record the data.

NOV 20–23

Model on the chip

Train, shrink to int8 and run it live on the board.

NOV 24–26

Tune and film

Play-test, measure and record the demo video.

Oliver: firmware. Eli: recording and training. Everyone records data. Live demo Dec 8.



RISKS

What could go wrong, and our fallback

If this goes wrong	We fall back to
The model runtime is too big for the chip	A lighter runtime, or a hand-written int8 network
Controller and data link will not share one USB port	Pick controller or recording mode at power-on
The camera slowly drifts	Hold still for 2 seconds to recalibrate
We run out of time	Fewer gestures, and a real keyboard for walking

Two things we will not cut: the model running on the chip, and the live demo.



On Dec 8, someone in this room plays Minecraft with it and grins.

Questions?

